Thomas Szwaja | Mathematik 1

Hausaufgabe Woche 5

Tutorium: Poker-Chip | Wintersemester 2025/26

Aufgabe 1 (WWT):

Lösung:

E_t	P_{t-1}	Q_{t-1}	P_t	Q_t	**Kommentar**
0	0	0	0	0	
0	0	1	0	1	
0	1	0	1	0	
0	1	1	1	1	
1	0	0	1	0	Achtung, $P_{t-1} \neq Q_{t-1}$
1	0	1	1	0	
1	1	0	0	1	
1	1	1	0	1	Achtung, $P_{t-1} \neq Q_{t-1}$

Aufgabe 2 (Schaltnetz):

Lösung:

Wir erinnern uns an die WWT für einen RS-Flipflop.

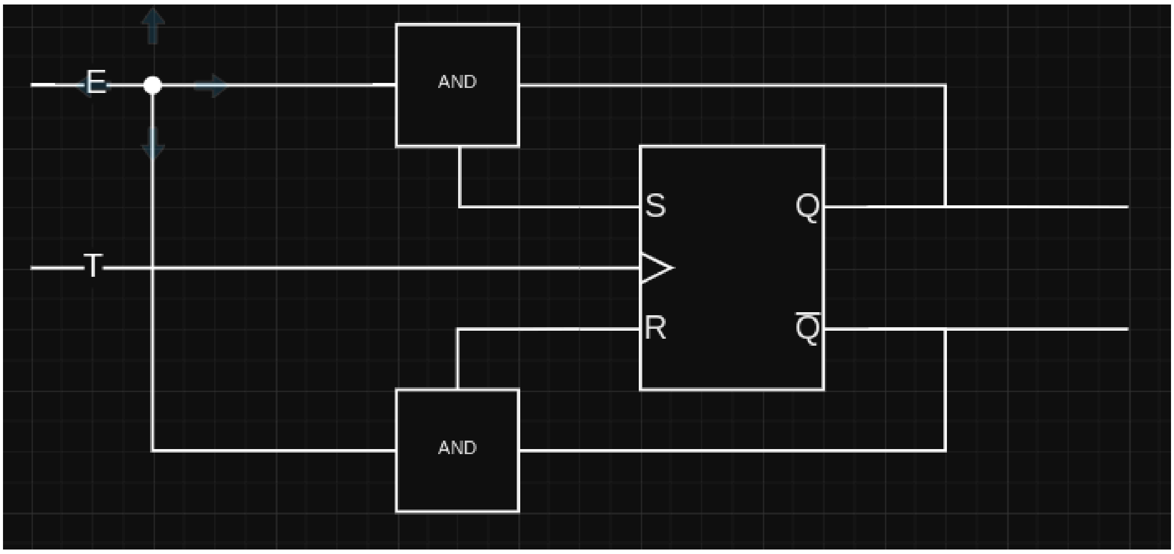
S_t	R_t	P_{t-1}	Q_{t-1}	P_t	Q_t
0	0	a	$\neg a$	a	$\neg a$
0	1	a	$\neg a$	1	0

S_t	R_t	P_{t-1}	Q_{t-1}	P_t	Q_t
1	0	a	$\neg a$	0	1

Wir nehmen an, wir brauchen eine Funktion in der Form $f : (E_t, Q_t) \to (S_t, R_t)$ so, dass die Eingaben (S_t, R_t) den entsprechenden Ausgang Q_{t+1} des RS-Flip-Flops verursachen. Wir fassen in einer Tabelle zusammen.

E_t	Q_t	Gewünschtes Q_{t+1}	Erforderliche S, R
0	0	0	$(0, 0)$
0	1	1	$(0, 0)$
1	0	1	$(1, 0)$
1	1	0	$(0, 1)$

Wir sehen daran, dass $S = E_t \wedge \neg Q_t$ und $R = E_t \wedge Q_t$. Wir sind bereit, den Schaltkreis zu entwerfen.



Aufgabe 3 (Eingangsflipflop):

Lösung:

Eingabeflipflop = Input Flip-Flop A flip-flop designed to accept and store external input in a larger sequential system. Think: keyboard input, sensor data, user button press - anything coming from outside your circuit into your state machine. What Makes a Good One? The question is asking you to think about desirable properties for this role:

Deterministic behavior - predictable state transitions No forbidden states - can't brick your system with bad inputs Ease of control - simple input interface Flexibility - can handle different input patterns (set, reset, toggle, hold)

ANSWER: the jk flip flop is based on the rs flip flop and what seems to be a significant difference, is that we can set state independently from previous state. we can say store a 1 and it does it no matter what. the toggle can only switch states.

another issue might be that the toggle could be harder to control because you really have to get things done in 1 clock sycle. if you fail to switch off E (the input) in the next clock cycle, you've toggled yourself back to your initial state. i'm not sure this exact problem exists, although it seems to be the case with the 1, 1 state in the jk flip-flop.

so the toggle ff seems to tick a lot of boxes: deterministic, non-brickable, simple interface. but it' works the best with toggling and holding, the other functions would require adjustments at least.